Welcome to IDS 701!

Exercise 1

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You are all enrolled in a Data Science Master's Degree. But what is Data Science?

In your group, please:

1. Define "data science"
2. Tell us what you hope to be able to do when you graduate (that you couldn't do before)?

What is data science (in this class)?

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Study of how to solve problems

What is data science (in this class)?

Study of how to solve problems by answering questions

What is data science (in this class)?

Study of how to solve problems by answering questions using quantitative methods.

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All data science tools can be viewed as methods of answering questions about the world (as embodied in your data).

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$$\underset{a \in \text{possible answers}}{\operatorname{argmin}} \quad \text{Loss Fn}(a)$$

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True in both a narrow technical sense:

$$\underset{a \in \text{possible answers}}{\operatorname{argmin}} \quad \text{Loss Fn}(a)$$

Also true in a more subtle but important sense we’ll discuss more later.

Solving Problems with Data Science

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     $\rightarrow$ Useful for automation (how would a human rate this job applicant?)

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     $\rightarrow$ Useful for automation (how would a human rate this job applicant?)
   - Causal Questions: The first half of this class!

Exercise 2: Causal Questions

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In your group, please:

1. Tell us what types of problems we can solve by answering causal questions?
2. Why is prediction (e.g., supervised machine learning or predicted values from regressions) not enough?

Causal Questions

Answering causal questions entails predicting the likely consequence of taking an action.

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Answering causal questions entails predicting the likely consequence of taking an action.

When we take an action, we are changing the data generating process (the underlying process that generates whatever data we may be feeding into our model). This disruption means that correlations that were present in the data previously may no longer apply.

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You collect data on hotel prices and occupancy rates for the past few years.

What do you think you'd find?

Why is Prediction Not Enough?

Higher prices would be correlated with higher occupancy rates!

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Do you think raising *your* prices would increase *your* occupancy rates?

Why is Prediction Not Enough?

Higher prices would be correlated with higher occupancy rates!

Do you think raising *your* prices would increase *your* occupancy rates?

Why is this past pattern not holding?

In the historic data, prices are high because something external was driving up demand. But that is no longer true if you raise your rates unilaterally.

Why is Prediction Not Enough?

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Why is Prediction Not Enough?

Correlation does not necessarily imply causation.

Causal inference is about understanding when correlation does imply causation, and why.

Class Structure

1. Causal Questions:
    - Theory of Causal Inference
    - Experiments
    - Non-Experimental (Observational)
2. Project Design (Solving Problems):
    - Working with Stakeholders
    - Types of Questions

Class Structure

1. Largely Flipped:
    - Read Before Class
    - Quizzes
2. Website: ds4humans.com

Laptop Policy

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We have very clear, rigorous evidence that:

- Even just the *temptation* to jump to social media, etc. is distracting.
- Laptops don't just hurt your learning — they harm the learning of those around you.
- Taking notes by hand is harder and slower... and that makes it work better.

Statistical Decision-Making & Uncertainty

Nick Eubank

Frequentist Hypothesis Testing

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Binary Decision Making: Reject Null Hypothesis if $p <$ threshold.

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2. Binary. But "surely, God loves the .06 nearly as much as the .05."

Frequentist Hypothesis Testing

Binary Decision Making: Reject Null Hypothesis if $p <$ threshold.

1. "No effect" is a strawman
2. Binary. But "surely, God loves the .06 nearly as much as the .05."
3. Conflated: depends on sample size and effect size.

Frequentist Hypothesis Testing

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- H_0 of zero effect is a straw man.
- Consequently, there will always exist a sample size that will allow you to reject the null hypothesis of zero. Always.

So clearly we can't just rely on hypothesis testing.

Evaluating Evidence

Empirical research generates evidence.

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Interpretation of evidence requires judgment.

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Interpretation of evidence requires judgment.

So how can we evaluate results more holistically?

Frameworks

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Bayesian:

- θ is a random variable.
- We are trying to estimate the distribution of likely values of θ.
- Explicitly accounts for prior beliefs.

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   - Frequentist.
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2. Maximum Likelihood Estimation
   - Still frequentist.
   - Gives estimate of "most likely" value.
   - Asymptotically, gives sampling distribution of estimate $Pr(\text{data}|\theta)$.
3. Bayesian
   - Matches intuitive idea: estimating $Pr(\theta|\text{data})$.

Confidence Intervals

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Matching confidence intervals given in regression.

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Technically no — it is, or it isn't.

Before you collect data, you can say there's a 95% chance you'll draw data that generates an estimate with a confidence interval that includes true θ.

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Technically no — it is, or it isn't.

Before you collect data, you can say there's a 95% chance you'll draw data that generates an estimate with a confidence interval that includes true θ. But...

Bernstein-von Mises Theorem

The distribution of the Maximum Likelihood Estimator ($Pr(data|\theta)$) converges to the Bayesian posterior distribution $Pr(\theta|data)$ as $n \to \infty$ given "reasonable" priors.

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The distribution of the Maximum Likelihood Estimator ($Pr(data|\theta)$) converges to the Bayesian posterior distribution $Pr(\theta|data)$ as $n \to \infty$ given "reasonable" priors.

So in small sample you have to be careful, but treating confidence intervals as if they are the distribution of the likely true value of a coefficient is defensible.

Bernstein-von Mises Theorem

How fast does MLE Sampling Distribution converge to Bayesian Posterior Distribution?

Depends, but in general faster when priors are weak (meaning before you ran your analysis, you would put similar likelihood on all possible values of θ).

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- The stronger your priors, the slower the convergence.

Bernstein-von Mises Theorem

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Priors is information that sets Frequentist and Bayesian stats apart.

But eventually, enough data will always overwhelm your priors.

At that point, Frequentist and Bayesian begin to converge.

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    - Binary, conflates magnitude and precision.
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    - *But* it converges to Bayesian as $n \to \infty$
3. Bayesian
    - What you're thinking of intuitively.

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In large samples, treating confidence intervals as if they are the distribution of the likely true value of a coefficient is defensible.

Caveats:

- Assumes MLE properly specified (ignores model uncertainty).
- Not true in small samples!

Observational Data & Regressions

Nick Eubank

Observational Data

No baseline differences: $E(Y^0|D=1) = E(Y^0|D=0)$

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Observational Data

No baseline differences: $E(Y^0|D=1) = E(Y^0|D=0)$

- Experiments: ensure with Law of Large Numbers

If we can't randomize, then we don't get to rely on LLN. So then what?

Welcome to observational data!

Observational Data

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Suppose also that we understand the source of those differences!

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Can we *adjust* for those differences?

Observational Data

Suppose there *are* baseline differences.

Suppose also that we understand the source of those differences!

Can we *adjust* for those differences?

$E(Y^0|D = 1, X) = E(Y^0|D = 0, X)$

That's what causal inference with observational data is all about.

Omitted Variable Bias

We want to know the effect of D_i on Y. Suppose the causal relationship is:

$$Y_i = \alpha^L + \beta^L D_i + \gamma A_i + e_i^L \tag{1}$$

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Omitted Variable Bias is the difference between the true coefficient β^L and the coefficient we estimate β^S.

Omitted Variable Bias

$$
\begin{aligned}
Y_i &= \alpha^L + \beta^L D_i + \gamma A_i + e_i^L \\
Y_i &= \alpha^S + \beta^S D_i + e_i^S
\end{aligned}
$$

So what is $\beta^L - \beta^S$?

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So what is $\beta^L - \beta^S$?

$$\beta^L - \beta^S = \text{relation between } A_i \text{ and } D_i * \text{ Effect of } A_i$$

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\beta^L - \beta^S &= \text{relation between } A_i \text{ and } D_i * \text{ Effect of } A_i \\
\beta^L - \beta^S &= \pi_1 * \gamma
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Where π_1 comes from estimating $A_i = \pi_0 + \pi_1 D_i + \mu_i$

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Asked for reason people with different Y^0 would sort differentially into $D=0$ and $D=1$.

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Remember our 2x2 discussion of baseline differences $E(Y^0|D=1) = E(Y^0|D=0)$?

Asked for reason people with different Y^0 would sort differentially into $D=0$ and $D=1$.

In other words, we needed an unobserved factor (A_i) that was correlated with D ($\pi_1 \neq 0$) and Y^0 ($\gamma \neq 0$).

Functional Forms & Matching

Nick Eubank

Functional Forms

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You work for a mortgage lender building a credit-risk model.

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You fit the following model:

$$\begin{aligned} delinquency_i = \beta_0 \quad &+ \quad \beta_1 credit_score_i + \beta_2 income_i \\ &+ \quad \beta_3 interest_rate_i + \epsilon_i \end{aligned}$$

What functional form assumptions are implicit in this specification?

Functional Forms

$$
\begin{aligned}
delinquency_i = \beta_0 \quad &+ \quad \beta_1 credit_score_i + \beta_2 income_i \\
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Functional Forms

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- Borrower A's income rises from $20,000 to $30,000
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Does Borrower A's delinquency risk decrease by more, less or the same amount as Borrower B's risk?

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Interest rates rise by 1 percentage point.

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Matching & Functional Forms

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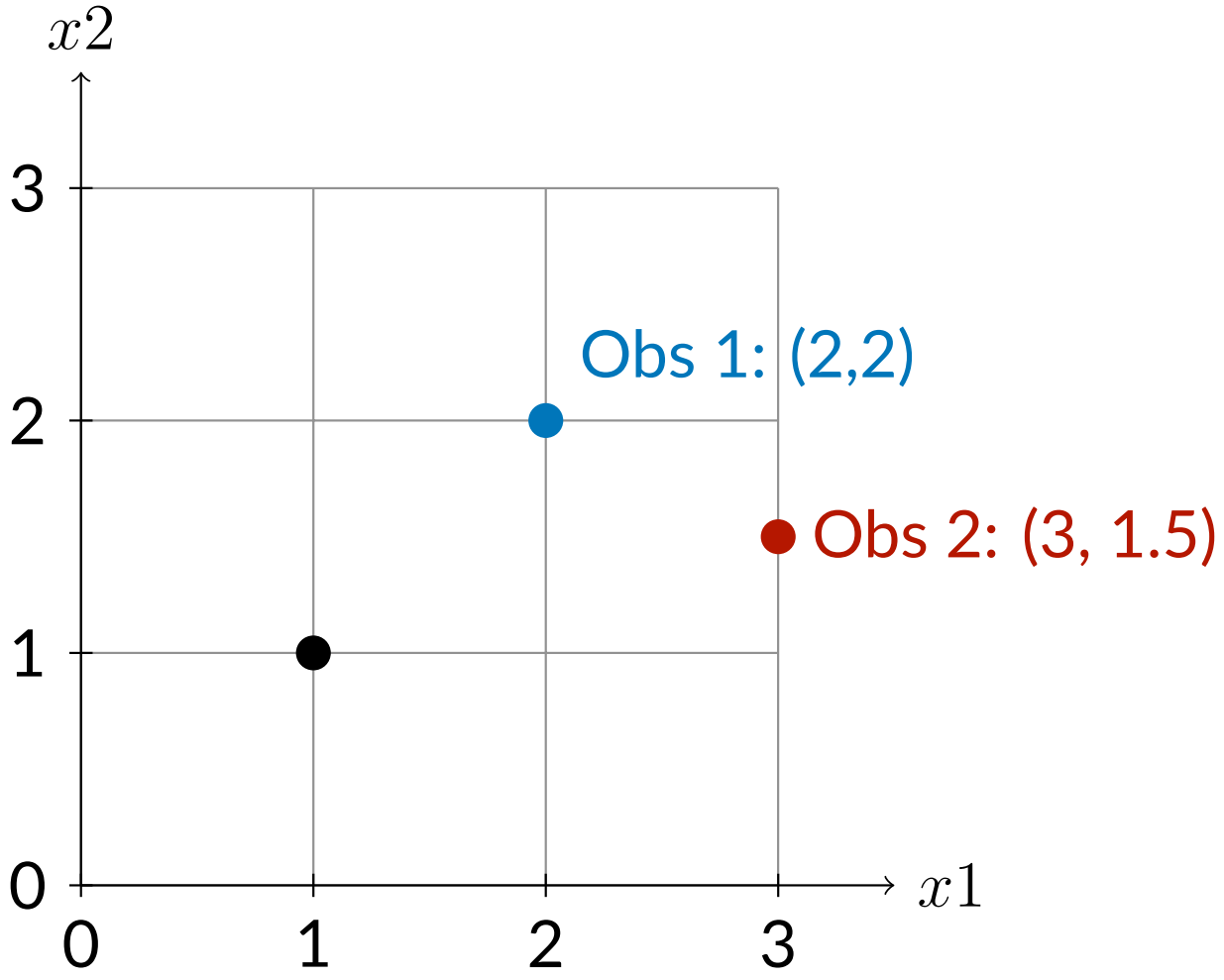
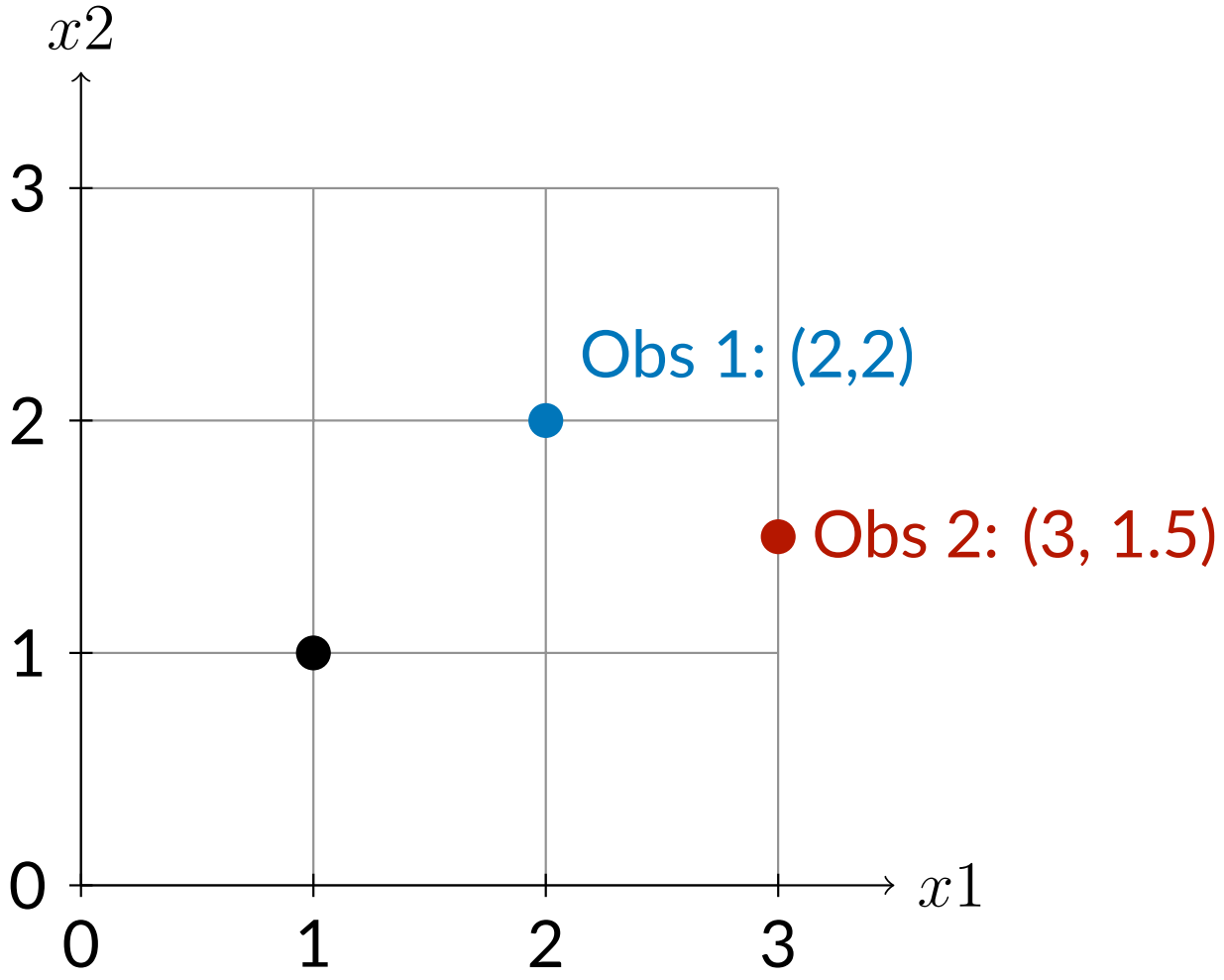
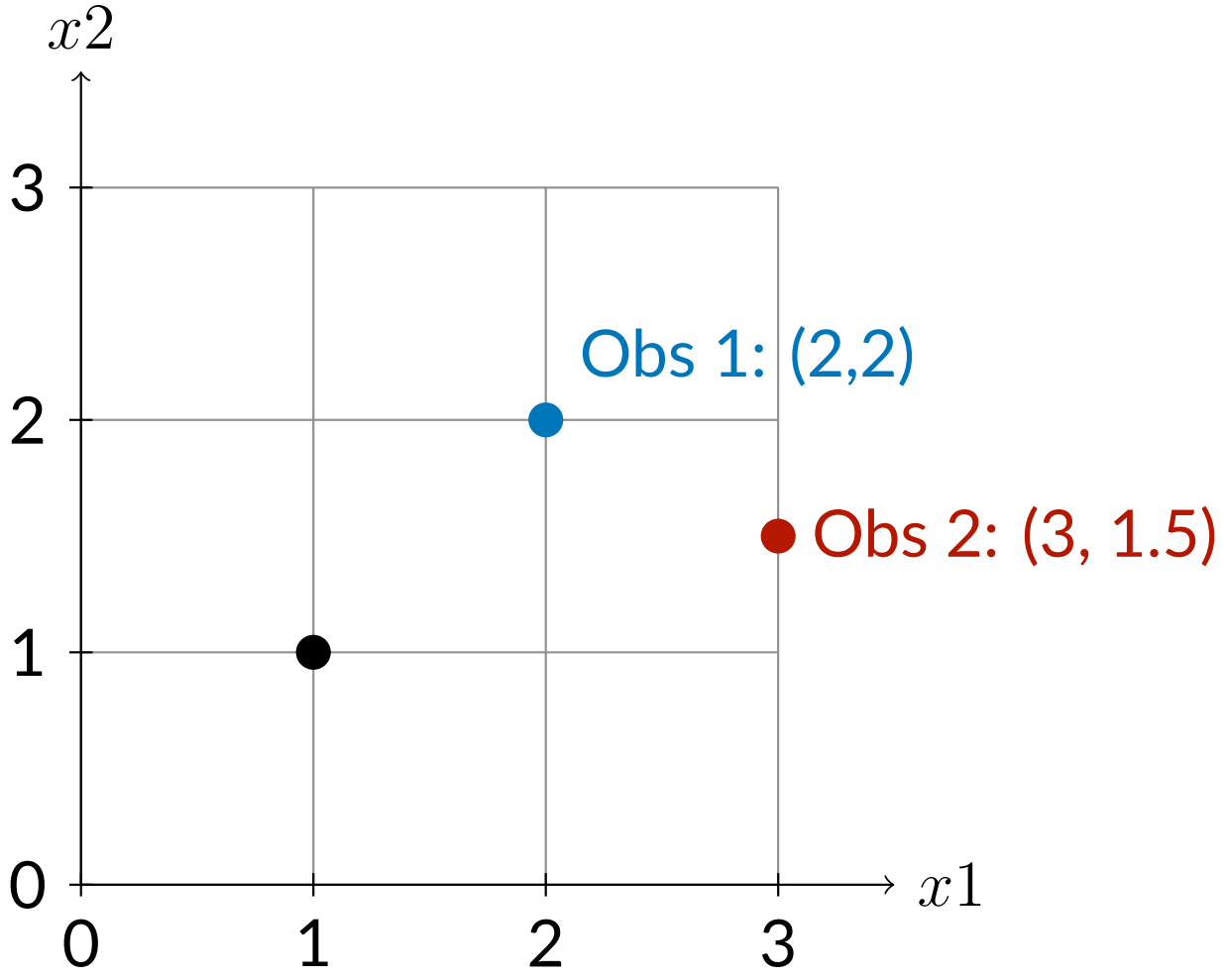
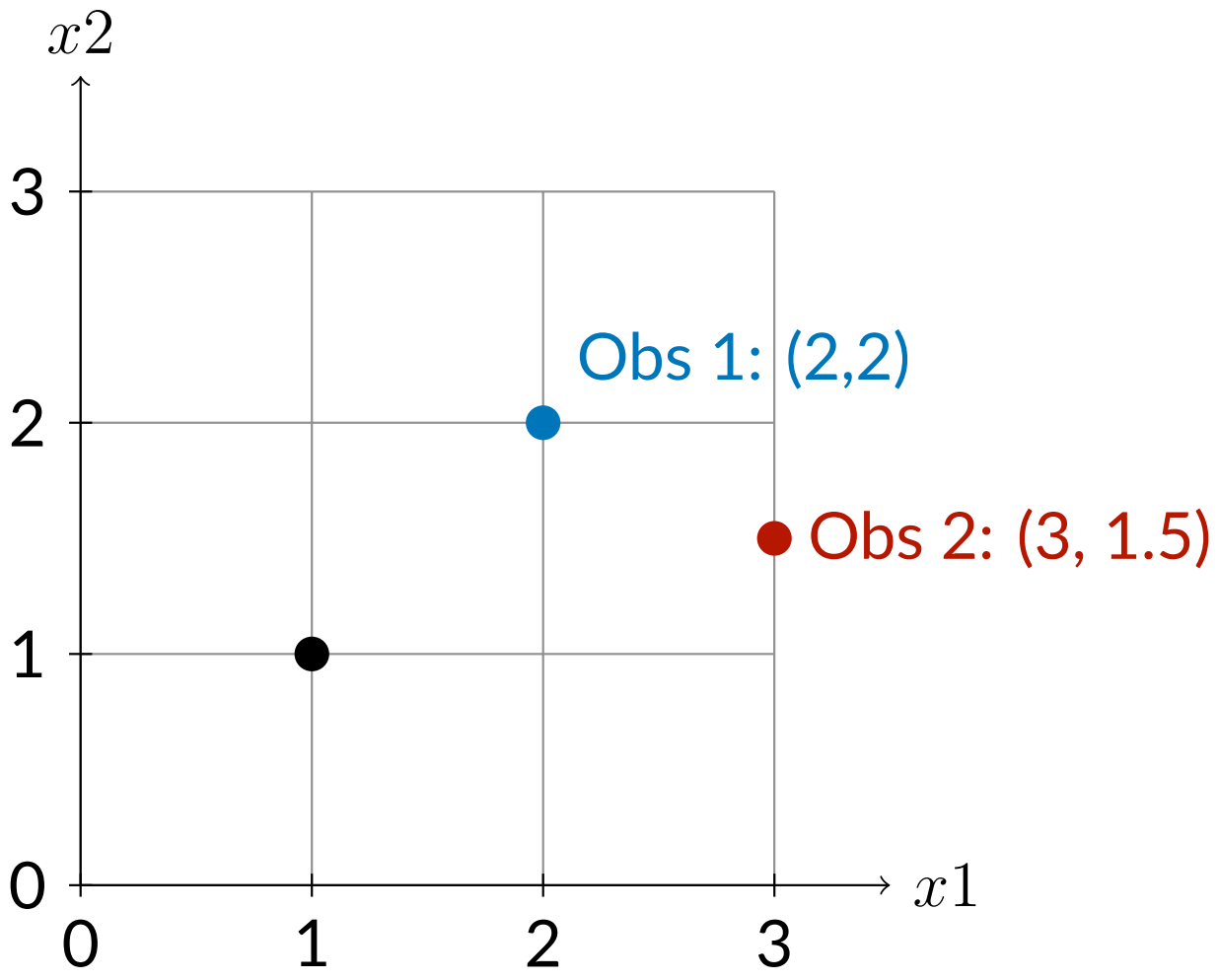
So is it a silver bullet for letting the data determine how our features impact outcomes?


$x2$
3
2
1
0
0
1
2
3
$x1$
Obs 1: (2,2)
Obs 2: (3, 1.5)



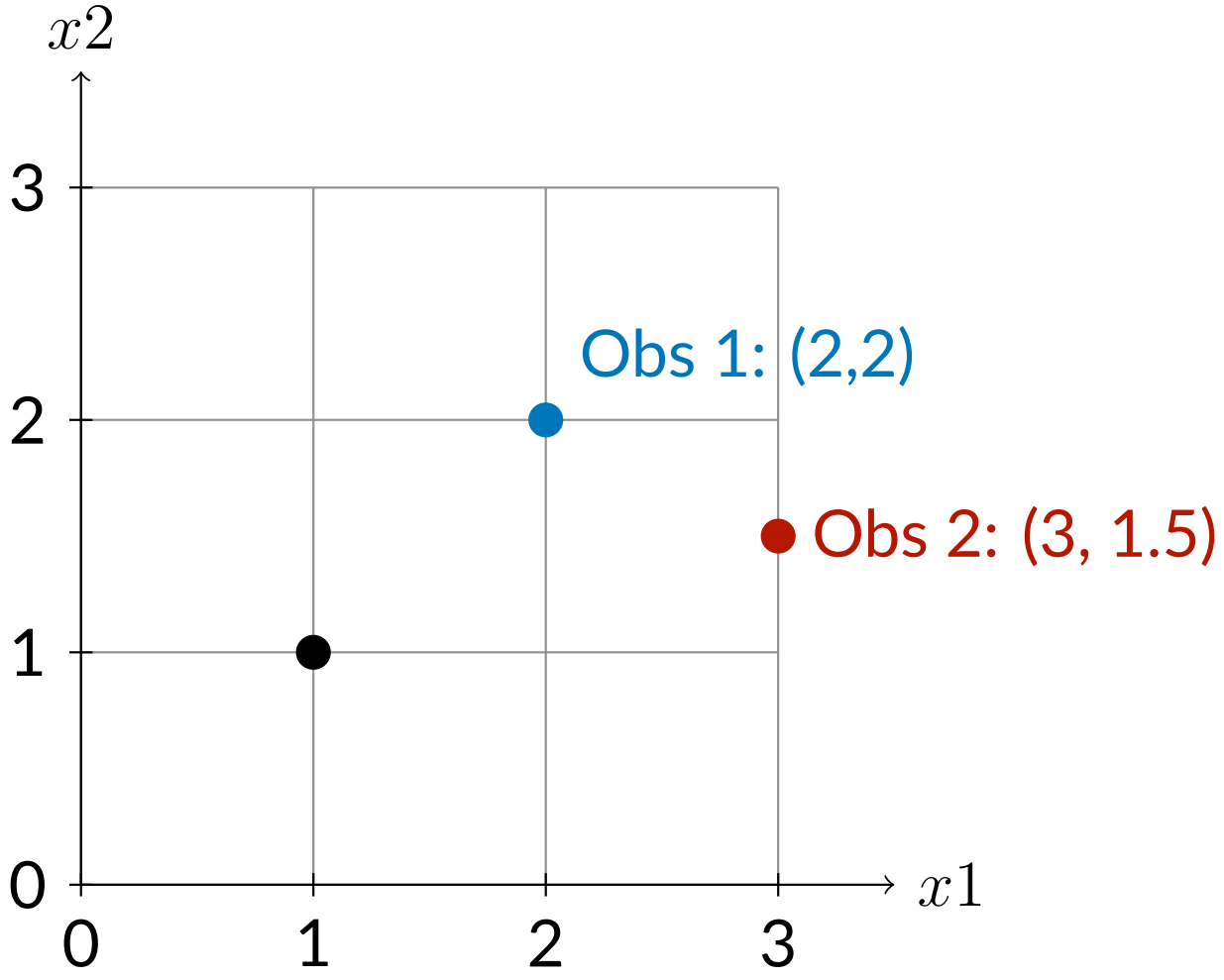
Which point should be matched with the Black point? Why?


$x2$
3
2
1
0
0
1
2
3
$x1$
Obs 1: (2,2)
Obs 2: (3, 1.5)



$\sigma_{x1} = \sigma_{x2} = 1$

Now which point?


$x2$
3
2
1
0
0
1
2
3
$x1$
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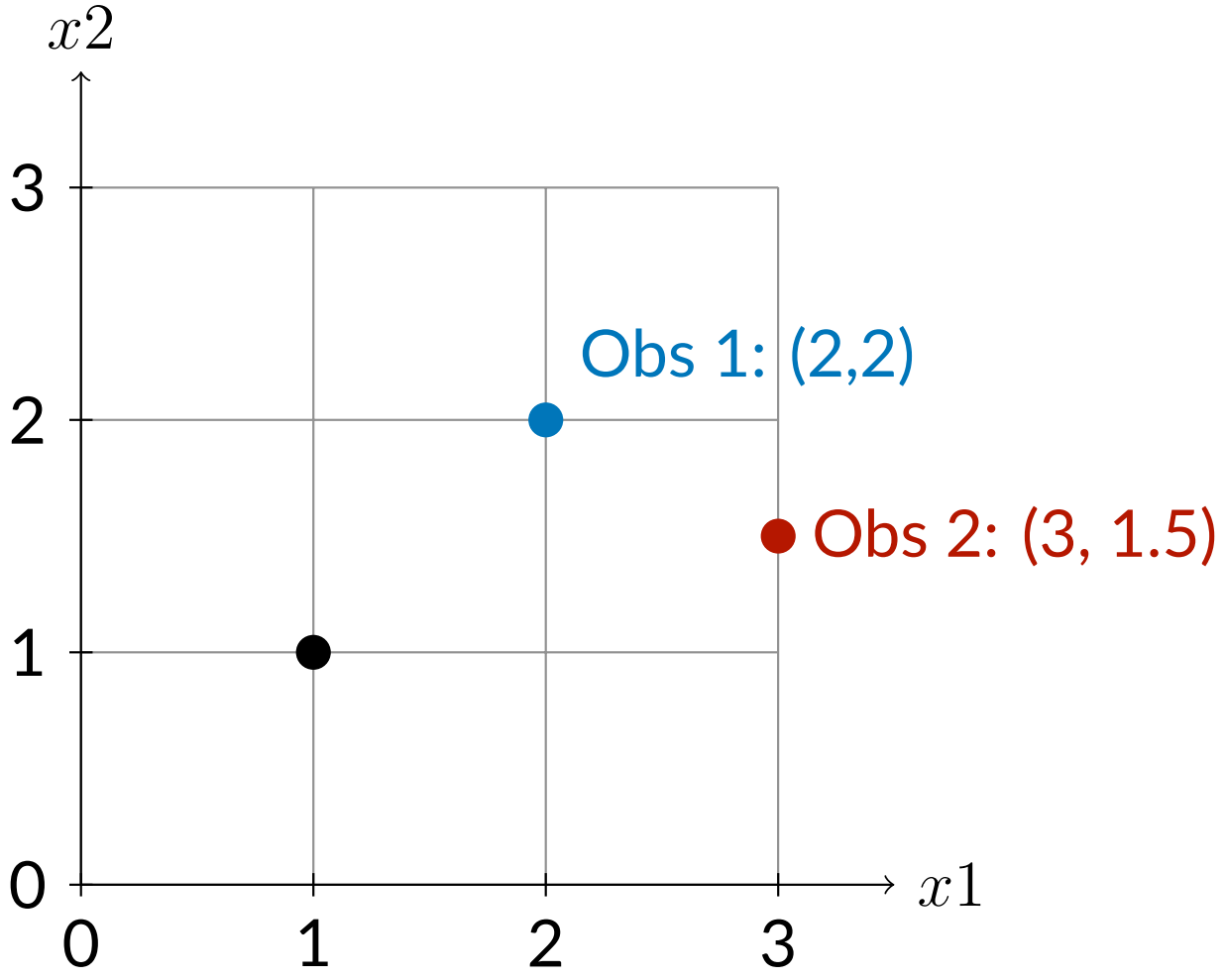
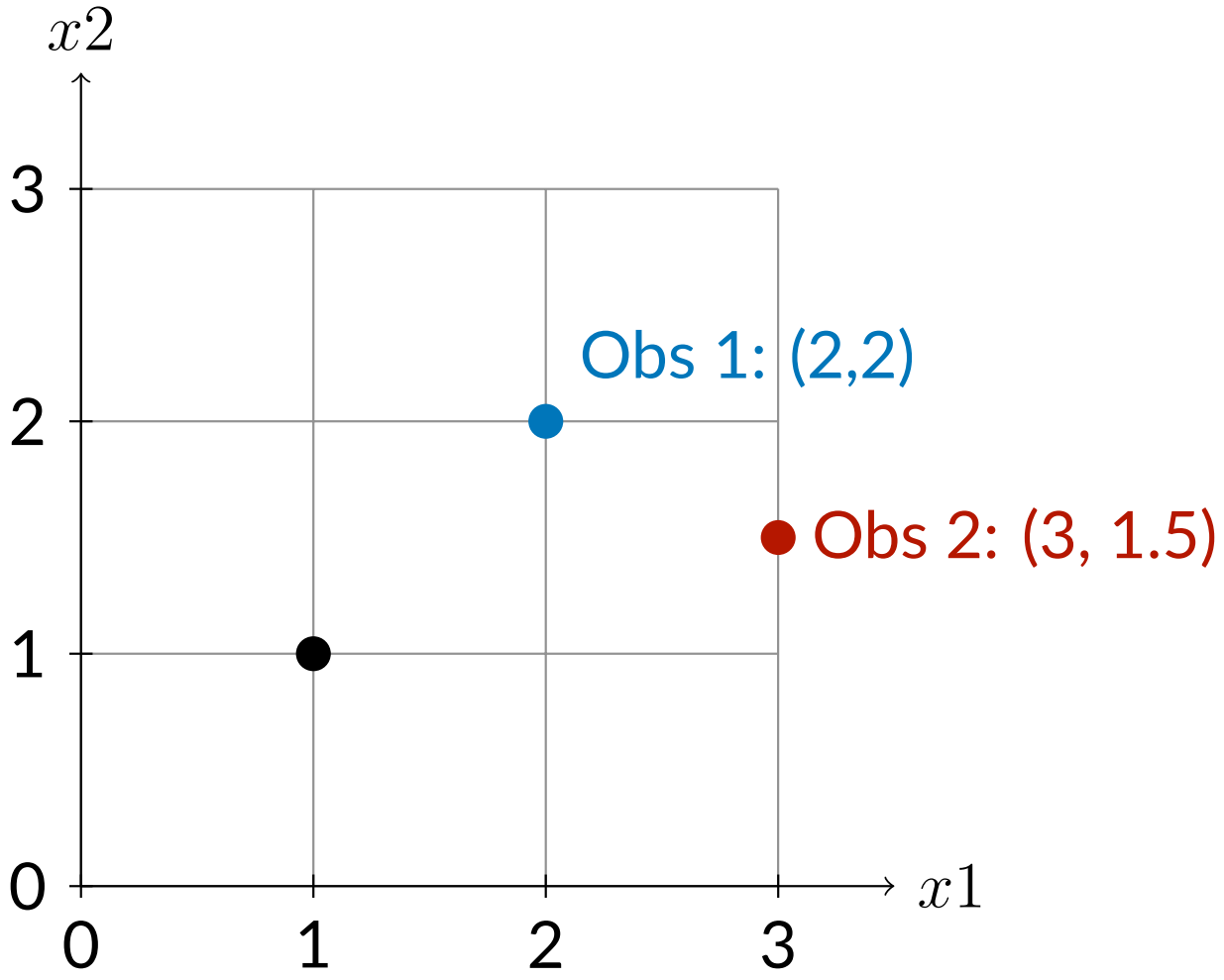


Suppose we add a new outlier data point to our dataset. We won't keep it when matching, but it means $\sigma_{x1} = 3$ while $\sigma_{x2} = 1$ still.

Now which point? (Hint: $\sqrt{(\frac{1}{3})^2 + 1^2)} > \sqrt{(\frac{2}{3})^2 + 0.5^2)}$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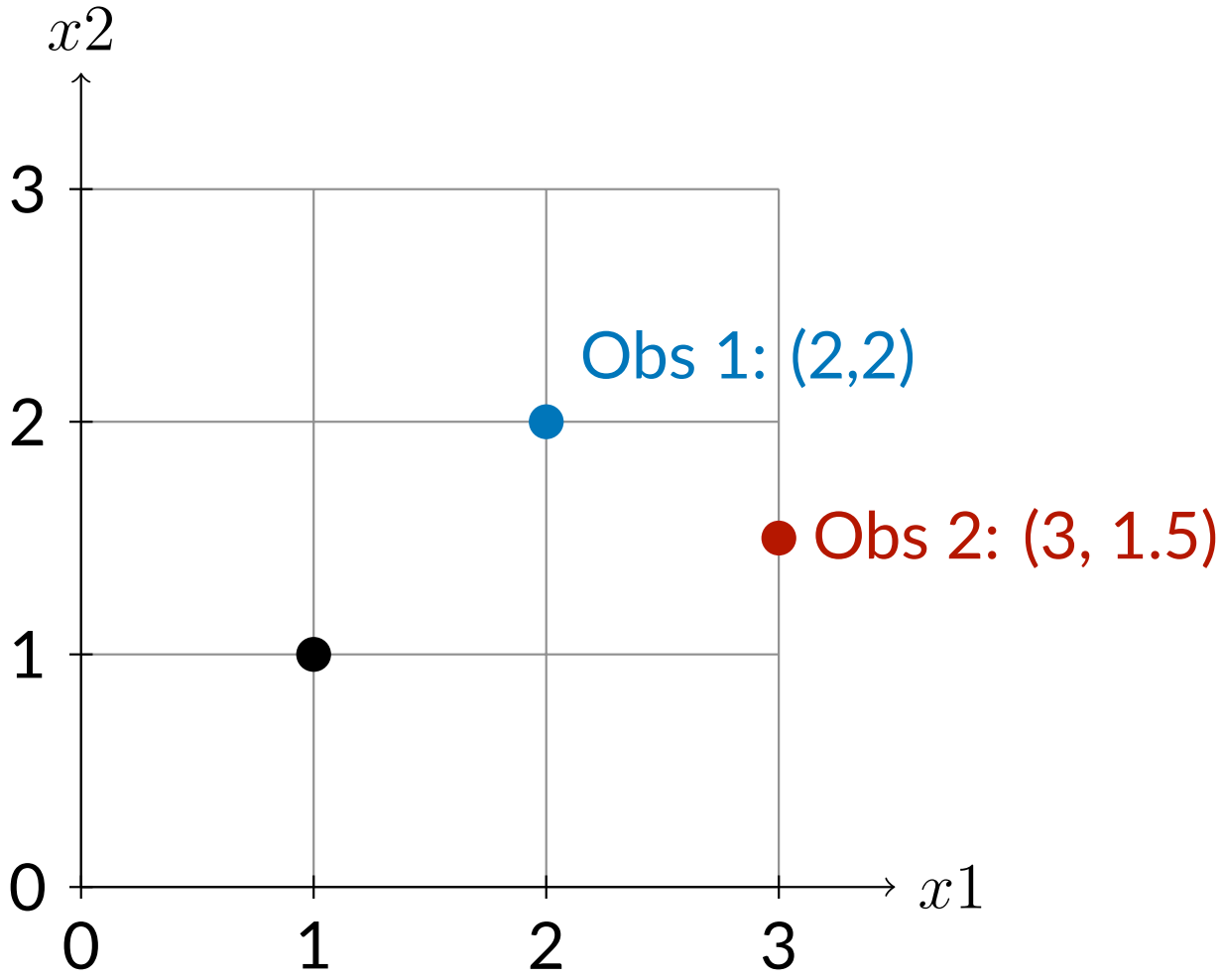
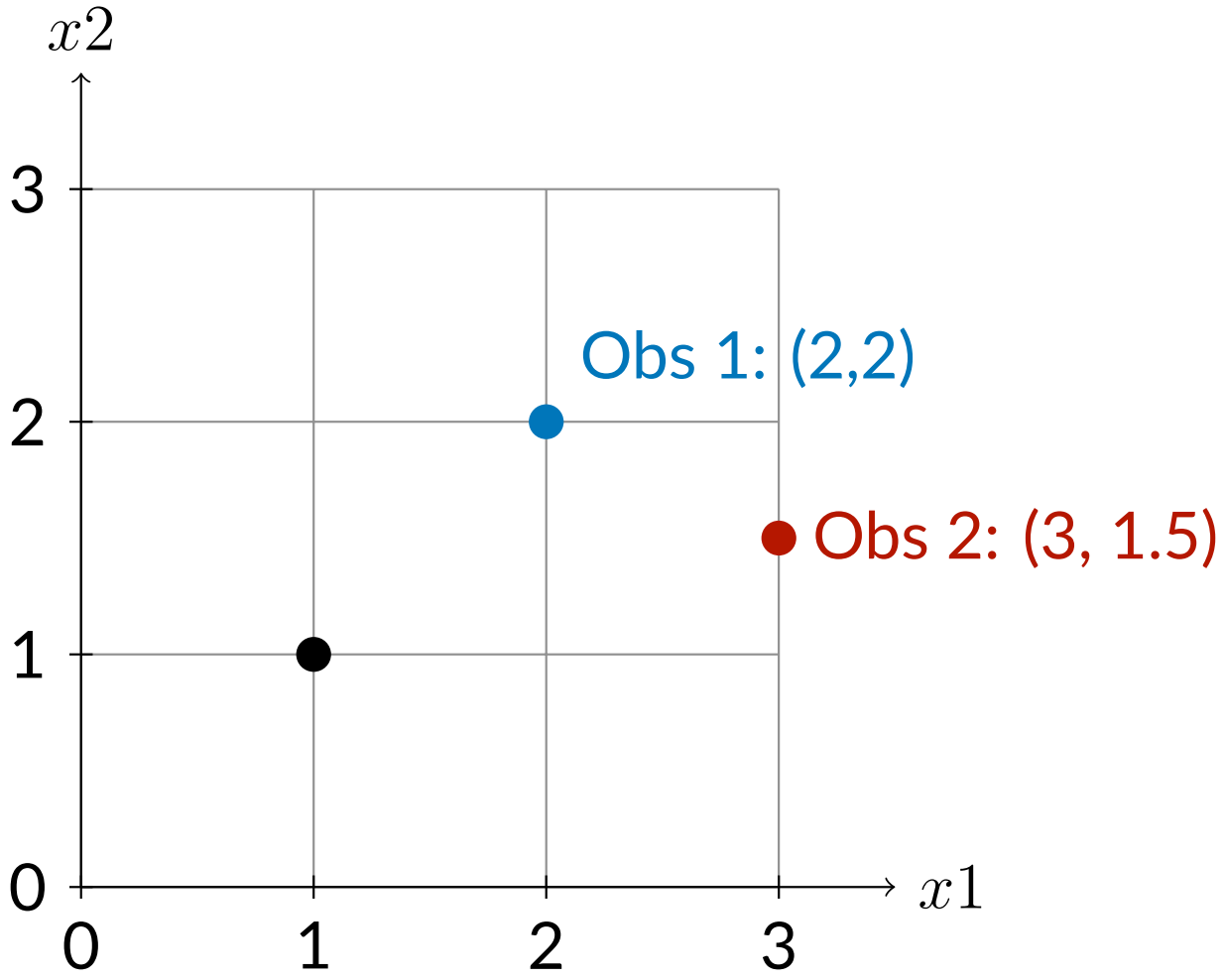


Our goal is to match observations with same potential outcomes. Does this new point impact how x impacts observation's potential outcomes?



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Then why did we let it shape the point to which we matched?

Matching and “Similarity”

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But you do still have to define your similarity measure!

And that requires choosing relative importance weights of different features, even if that isn’t explicitly obvious.

Curse of Dimensionality, Propensity Scores & Matching

Nick Eubank

Curse of Dimensionality

Assume we are doing Coarse Exact Matching on 200 observation dataset. 100 obs Treated, 100 obs control.

Assume two covariates, $x_1 \in 1, 2, 3, 4$ and $x_2 \in 1, 2, 3, 4$.

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We're matching, so probably *not* uniform, but prob fine.

Curse of Dimensionality

Assume we are doing Coarse Exact Matching on 200 observation dataset. 100 obs Treated, 100 obs control.

Assume three covariates, $x_1 \in 1, 2, 3, 4$, $x_2 \in 1, 2, 3, 4$, $x_3 \in 1, 2, 3, 4$.

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Sure to be plenty of empty cells, but also cells that match.

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Assume four covariates, $x_1 \in 1, 2, 3, 4$, $x_2 \in 1, 2, 3, 4$, $x_3 \in 1, 2, 3, 4$ and $x_4 \in 1, 2, 3, 4$.

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Oof... mostly empty cells?

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Assume five covariates, $x_1 \in 1, 2, 3, 4$, $x_2 \in 1, 2, 3, 4$, $x_3 \in 1, 2, 3, 4$ and $x_4 \in 1, 2, 3, 4$, $x_5 \in 1, 2, 3, 4$.

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Oof... Even if most are in an area of overlap, getting *real* hard.

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Even with just five covariates, each with only four discrete values (no continuous variables!), finding matches gets quickly fast!

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Thus the allure of propensity score matching.

Conditional Independence Assumption

As discussed before, our goal in observational causal inference is to find and condition on covariates X such that:

$$E(Y^0|D_i = 1, X) = E(Y^0|D_i = 0, X)$$

Statisticians generally use the slightly stronger Conditional Independence Assumption (which implies No Baseline Differences):

$$(Y^1, Y^0) \perp\!\!\!\perp D|X$$

So when you see $(Y^1, Y^0) \perp\!\!\!\perp D|X$, that implies no baseline differences (and also no differential treatment effects).

Propensity Scores

Propensity Score Theorem:

$$\text{if } (Y_i^1, Y_i^0) \perp\!\!\!\perp D_i | X$$
$$\text{then} (Y_i^1, Y_i^0) \perp\!\!\!\perp D_i | p(X_i)$$

Where $p(X_i)$ is the propensity score:
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Mathematical Proof in Angrist and Pischke, *Mostly Harmless Econometrics*, Sec 3.3.2. or `https://youtu.be/nnC1tlircdE`

Propensity Scores Matching

Why does conditioning on Propensity Score $\rightarrow$ No Baseline Differences?

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Matching on covariates X_i makes intuitive sense:

Pair T and C observations that look the same and thus (we hope) have as close to the same potential outcomes (Y_i^0, Y_i^1) as possible.

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Baseline differences are violated when a factor (X_i) is correlated with Y_i^0 and D_i.

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There's no Y_i^0 in that equation!

Baseline differences are violated when a factor (X_i) is correlated with Y_i^0 and D_i.

Propensity Score Matching just focuses on breaking correlation between factors X_i and D_i.

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Weakness:

1. You *aren't* trying to make pairs as comparable as possible in terms of Y_i^0, Y_i^1.
   - Thus sample left after matching may be unbalanced and inefficient.
2. Proofs assume you *know* $Pr(D_i = 1|X_i)$. Issues when estimating.

My (Opinionated) Summary

- If you have data to match on covariates, do.
- Propensity score matching is an OK choice if (a) you have a strong understanding of the true $Pr(D_i = 1|X_i)$ and (b) you aren't dropping too many observations (potentially giving rise to imbalance).

Potential Outcomes Recap

Nick Eubank

The Big Picture

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- We can measure:

  $$\widehat{ATE} = E(Y^1|D=1) - E(Y^0|D=0)$$

- If no baseline differences:

  $$\widehat{ATE} = ATT$$

- If also no differential treatment effects:

  $$\widehat{ATE} = ATT = ATE$$ 🥳

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We want to answer causal questions because we are considering taking an action at scale, and want to know the likely effect.

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- Change the price of a product.
- Change webpage design.

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- Change the price of a product.
- Change webpage design.

Average effect will be ATE.

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- Does NOT require no differences in Y across individuals.

Requires no average differences in outcome Y between $D = 0$ and $D = 1$.

No Differential Treatment Effects

$$E(Y^1|D=1) - E(Y^0|D=1) = E(Y^1|D=0) - E(Y^0|D=0)$$

The effect of T is the same (on average) for both groups.

No Differential Treatment Effects

$$E(Y^1|D=1) - E(Y^0|D=1) = E(Y^1|D=0) - E(Y^0|D=0)$$

The effect of T is the same (on average) for both groups.

- If not met, but no baseline differences is met, still get *a* treatment effect: ATT.

SUTVA & Indicator Variables

Nick Eubank

Potential Outcome Assumptions

Hiding in all that math is a big assumption:

Stable Unit Treatment Value Assumption (SUTVA)

- Treatment is homogeneous across units
- Treatment status of one unit does not effect treatment status of others.

SUTVA Exercise 1

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It's 2021, and the Covid Epidemic is in fully swing. Duke wants to test a new covid vaccine.

They propose give half of Duke students the vaccine and half a placo.

They then intend to measure differences in covid positive rates over the following 3 months.

- Why does this violate SUTVA?
- How could you fix the problem?

SUTVA Exercise 1: Answers

- Why does this violate SUTVA?

SUTVA Exercise 1: Answers

- Why does this violate SUTVA?

  Duke students who receive placebo are at less risk of covid because half of their friends are no longer getting sick! By making control and treatment look more similar than they would if everyone was treated or if everyone was untreated, the estimated effect of the vaccine will be lower than the true ATE.

SUTVA Exercise 1: Answers

- Why does this violate SUTVA?

  Duke students who receive placebo are at less risk of covid because half of their friends are no longer getting sick! By making control and treatment look more similar than they would if everyone was treated or if everyone was untreated, the estimated effect of the vaccine will be lower than the true ATE.
- How could you fix the problem?
  - Treat (or don't treat) entire dorms.
  - Give it to a much smaller proportion of students.
  - Don't use Duke students — use people from Durham.

SUTVA Exercise 2

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They run an experiment where 10% drivers in Durham experience this higher pay model.

The outcome they measure is overall driver revenue.

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- Why does this violate SUTVA?
    - The higher pay may cause more drivers to be out on the road from the treatment group.
    - This will make it harder for control drivers to pick up rides.
    - But also: it won't increase people on the road as much as the new rate would if it were offered to all drivers, so the drivers with higher pay will face *less* competition for rides than if everyone had the treatment.

SUTVA Exercise 2: Answers

- Why does this violate SUTVA?
    - The higher pay may cause more drivers to be out on the road from the treatment group.
    - This will make it harder for control drivers to pick up rides.
    - But also: it won't increase people on the road as much as the new rate would if it were offered to all drivers, so the drivers with higher pay will face *less* competition for rides than if everyone had the treatment.
    - So this would increase the apparent effectiveness of the policy.

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  - Assign entire cities to control or treatment. Compare cities.
  - Randomize across days or weeks.

SUTVA Exercise 3

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Instagram is testing a change to their feed algorithm.

The treatment is that for 10% of users, their feeds will only show happy content.

They outcome they measure is whether these users spend more time on Instagram than users with the current feed algorithm.

- Why does this violate SUTVA?
- How could you fix the problem?

Indicator Vars: Does Applicant Get Loan? (Binary Outcome)

	coef	std err	t	P>|t|
Intercept	0.4921	0.050	9.764	0.000
has_college	0.1020	0.010	9.962	0.000
poc	-0.2735	0.011	-24.398	0.000
has_college:poc	0.2910	0.018	16.381	0.000
female	0.0087	0.008	1.057	0.291
age	-0.0006	0.002	-0.261	0.794
age_squared	7.428e-06	2.26e-05	0.328	0.743

1. How much more likely is a White college grad to get a loan than a White non-college grad? ("poc" = "person of color")
2. How much more likely is a non-White college grad to get a loan than a non-White non-college grad?
3. How much more likely is a non-White college grad to get a loan than a White non-college grad?

Experiments and Internal Validity

Nick Eubank

Experiments & Causal Assumptions

1. No baseline differences

   Very likely (known probability) given randomization & law of large numbers.

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   Very likely (known probability) given randomization & law of large numbers.

3. SUTVA / No Interference:

   Choice of randomization units.

   Requires critical reasoning/domain expertise.

Experimental Internal Validity Checks

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2. Sample ratio: does number of people in treatment and control match expectation given randomization?
3. A/A test: checks whether testing infrastructure induces differences.

Experiments and External Validity

Nick Eubank

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Do the results *generalize*.

External validity can depend on many things:

- Are the entities in the study drawn from the same population in the new setting?
- Is the treatment the same in the new setting?
- Is the broader context the same in the new setting (artificiality)?

External Validity: Sample Composition

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Psychology studies mostly run on Western, Educated students from Industrialized, Rich, and Democratic countries (they're WEIRD).

External Validity: Treatment Differences

A new car dealer sends out mailers offering $3,000 off list.

Customers who receive more likely to come buy car in following 6 months.

External Validity: Treatment Differences

A new car dealer sends out mailers offering $3,000 off list.

Customers who receive more likely to come buy car in following 6 months.

Will that generalize to:

- $1,000 discount?
- Mailers from used car dealership?
- Coupons handed out by hand?

External Validity: Artificiality

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(One explanation for medical study issues)

- Hawthorne Effect: People behave differently when know observed.

  Originally documented in studies of workers.

External Validity

But enough of the easy causes…

External Validity: Exercise 1

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- Do you think free samples would have the same effect for Bud Light?

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Distinguishes internal validity from external validity.

External Validity: Exercise 2a

Apple offers a $300 discount on laptops in a randomly selected set of cities for 24 hours.

Sales on day of the sale so good (compared to sales in other cities), Apple considering lowering list prices.

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- Do you think sales would increase the same amount if Apple reduced prices everywhere?

External Validity: Exercise 2b

A national used car dealership chain offers $1,000 discount at locations in a random set of cities for 24 hours.

Sales on day of the sale so good (compared to sales in their other locations), company considering lowering list prices.

External Validity: Exercise 2b

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- Do you think sales would increase the same amount if the dealer reduced prices everywhere?

External Validity: Time-shifting & Competition

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Relative magnitudes depend on produce and market structure.

External Validity: Exercise 3a

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A/B testing shows that because users don’t get the “right” result as quickly, they tend to search more and thus see more ads, increasing revenue!

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- If you ran Bing, would you adopt this new algorithm?

External Validity: Exercise 3b

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- If you ran Google, would you adopt this new algorithm?

External Validity: Exercise 3b

Google has launched a new algorithm that makes search results slightly worse.

A/B testing shows that because users don't get the "right" result as quickly, they tend to search more and thus see more ads, increasing revenue!

- If you ran Google, would you adopt this new algorithm?
- Guess what? They did! (According to a lot of docs and reporting from google trial)

External Validity: Exercise 4

Google is revising its algorithm (to actually help customers).

Google has found good recipe sites (non-spam sites) tend to have additional, novel prose around the recipe.

A/B testing shows promoting these pages increases user searches for recipes (including searches for *new* recipes) over 1 month period.

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A/B testing shows promoting these pages increases user searches for recipes (including searches for *new* recipes) over 1 month period.

- Do you think this result is likely to hold if rolled out to all users over the long run? Why or why not?

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- Content policing: "unalive," "seggs"
- Surge pricing: drivers who coordinate to create apparent driver "shortages" by all signing off.

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    - Adversarial Users
5. Others?

Overall Evaluation Criteria & P-Values

Nick Eubank

P-Values

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.0821	0.040	2.053	0.040	0.004	0.160
C(education)[T.1]	-0.0017	0.057	-0.030	0.976	-0.113	0.110
C(education)[T.2]	-8.953e-05	0.042	-0.002	0.998	-0.082	0.082
C(education)[T.3]	-0.0025	0.039	-0.065	0.948	-0.079	0.074
C(education)[T.4]	-0.0047	0.038	-0.124	0.901	-0.080	0.070
black	-0.0316	0.008	-4.076	0.000	-0.047	-0.016
yearsexp	0.0032	0.001	3.665	0.000	0.001	0.005
computerskills	-0.0186	0.011	-1.616	0.106	-0.041	0.004
female	0.0112	0.010	1.165	0.244	-0.008	0.030

What is the probability that the *true* effect of reporting computer skills on your resume is 0 and that the difference we see between resumes with and without reported computer skills is the result of chance variation?

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And doesn't that *feel* like what a p-value is?

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probability we would observe a coefficient at least this large *in a world where the Null Hypothesis is true*

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Fudging a little:

$$Pr(H_0 \text{ is true}|\text{Data}) = \frac{Pr(\text{Data}|H_0\text{is true}) * Pr(H_0\text{is true})}{Pr(\text{Data})}$$